Aim

To observe the Gibbs phenomenon of CTFS of a periodic square wave using MATLAB.

Theory

The Gibbs phenomenon occours in Fourier series approximations of discontinous periodic signals (like square waves) where we notice over/undershoots and ripples near the jump discontinuties.

When the number of terms/samples approches infinity, we observe that the over/undershoots approach 9% of the height of the discontiunity.

To find the CTFS for a periodic square wave, we will first find Cn via:

$$C_n = \sum_{n=-\infty}^{\infty} \frac{2\sin\left(\frac{n\pi}{2}\right)}{n\pi}$$

Then we will find the function by using:

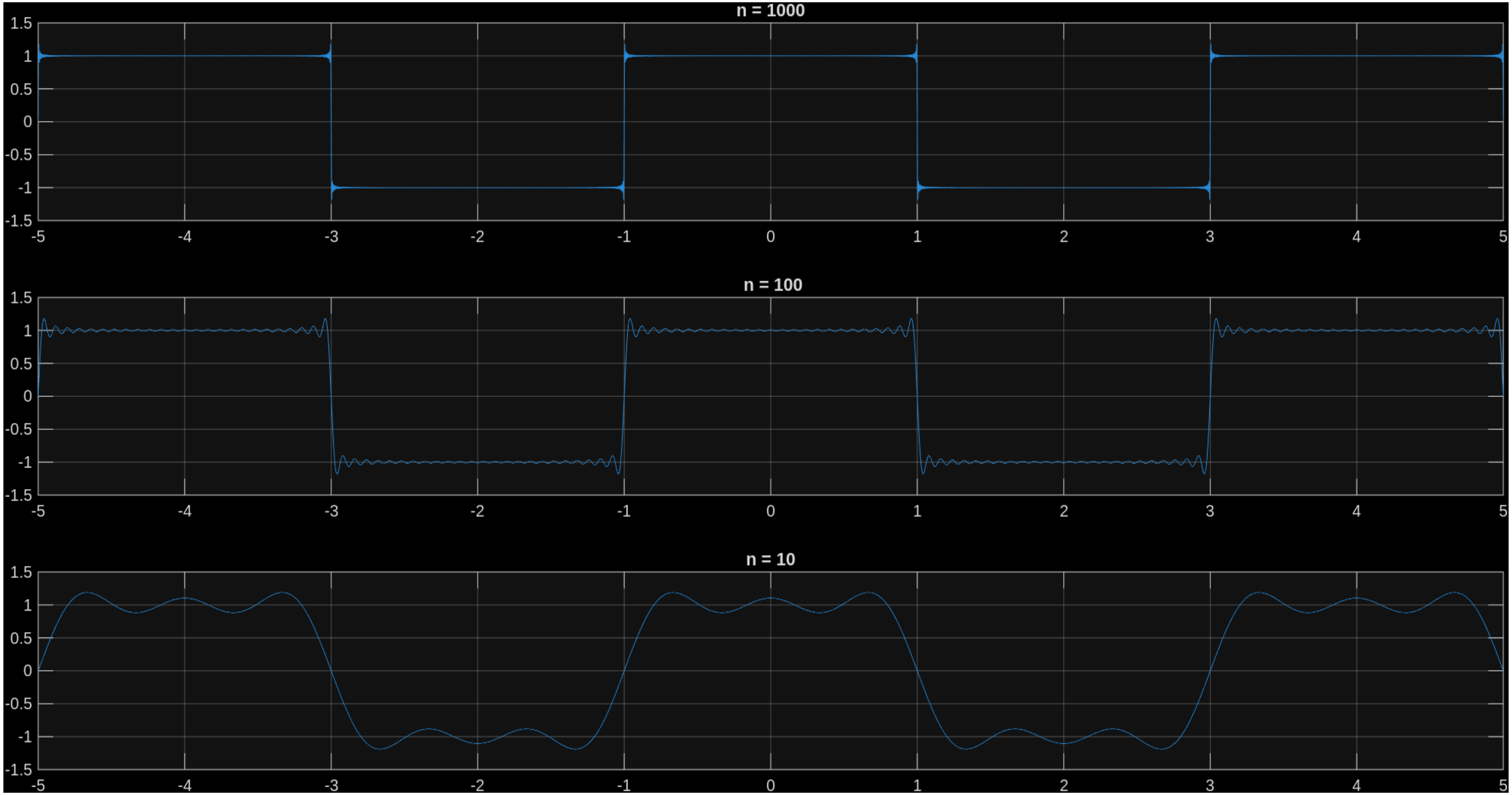
$$f(k) = C_n + e^{\left(jn\pi\frac{t}{2}\right)}$$

Where j = √-1

MATLAB Code

```
t = -5:10^-4:5;

f1 = zeros(1, length(t));
f2 = zeros(1, length(t));
f3 = zeros(1, length(t));

n = -500:1:500;
Cn = 2*(sin(n*pi/2))./ (n*pi);

for k = 1:1:length(n)
    if n(k) ~= 0
        f1 = f1 + (Cn(k) * exp(sqrt(-1) * n(k) * pi * (t/2)));
    else
        f1 = f1+0;
    end
end

n = -50:1:50;
Cn = 2*(sin(n*pi/2))./ (n*pi);

for k = 1:1:length(n)
    if n(k) ~= 0
        f2 = f2 + (Cn(k) * exp(sqrt(-1) * n(k) * pi * (t/2)));
    else
        f2 = f2+0;
    end
end

n = -5:1:5;
Cn = 2*(sin(n*pi/2))./ (n*pi);

for k = 1:1:length(n)
    if n(k) ~= 0
        f3 = f3 + (Cn(k) * exp(sqrt(-1) * n(k) * pi * (t/2)));
    else
        f3 = f3+0;
    end
end

figure;

subplot(3,1, 1)
plot(t, f1)
title('n = 1000')
grid on;

subplot(3,1, 2)
plot(t, f2)
title(" n = 100")
grid on;

subplot(3, 1, 3)
plot(t, f3)
title("n = 10")
grid on;
```

Results



Conclusion

In this experiment, we have demonstated the Gibbs phenomenon through CTFS analysis of a periodic square wave in MATLAB and have found a ~9 over/undershoot at jump discontinuities despite increasing the number of harmonics.